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Paleomagnetism of the Marble Bar Chert Member, Western Australia: Implications for apparent polar wander path for Pilbara craton during Archean time

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Abstract

The Archean Biosphere Drilling Project (ABDP) drilled a continuous 270 m long oriented core from the Towers Formation, which includes the Marble Bar Chert Member (3456.1–3476.0 Ma) in the Pilbara craton, northwestern Australia. A paleomagnetic study of 261 discrete specimens, collected from a 158.5 to 182.0 m section of the Marble Bar Chert Member, revealed two distinct magnetic components (LT and MT). The MT component yields seven different mean paleomagnetic directions clustered as MB1 to MB7. These, together with the published paleomagnetic poles of early Archean rocks from the Pilbara craton, draw a continuous paleomagnetic pole path, which likely to be regarded as the early to late Archean apparent polar wander path (APWP) for the Pilbara craton. The APWP implies that the Pilbara craton underwent a latitudinal drift of about 21° during the interval when the magnetization of the Marble Bar Chert Member was acquired. The estimated speed of the lateral drift is 12–112 cm/yr (120–1120 km/Myr), which is large compared with current plate motion velocities, suggesting that continents might have moved during the Archean faster than in the Phanerozoic.
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Keywords: paleomagnetism; Archean; Pilbara craton; Marble Bar Chert; apparent polar wander path

1. Introduction

The Pilbara craton, located in the northwestern part of Australia, is one of the important cratons because of the continuous geological record that yields a well-constrained apparent polar wander path (APWP) during Precambrian time. Idnurm and Giddings [1] reconstructed the APWP of the Pilbara craton from 3.5 to 0.6 Ga by compiling of published paleomagnetic poles. However, they adopted from McElhinny and Senanayake [2] only one paleomagnetic pole of the Pilbara craton before 2.8 Ga. Recently, Strik et al. [3] made a significant contribution to the reconstruction of the

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APWP for late Archean time. They studied 2775 to 2715 Ma rocks from the Nullagine synclinorium, East Pilbara Basin, and provided the latest reconstruction of the late Archean APWP, although early to middle Archean paleomagnetic records from the Pilbara craton remain scarce.

One of the main controversial topics of Archean dynamics is whether the plate tectonic process existed or not. In spite of a long list of researches addressing this topic, this question remains controversial. Plate tectonic models have been constructed from field observations for Archean greenstone terranes since the 1970s [e. g., 4,5]. Komiya et al. [6] reported the existence of an accretionary complex in the ca. 3.8 Ga Isua supracrustal belt, southern west Greenland, and claimed that the plate tectonics process has its roots in early Archean time. Kato and Nakamura [7] reported the geochemical signature of the early Archean cherts and documented that these chert were deposited in a variety of environments ranging from a mid-oceanic spreading center to a convergent plate boundary, suggesting the existence of the plate tectonics process in early Archean time. However, a contrary model (vertical tectonics largely controlling the Archean dynamics and ruling out plate tectonic processes during Archean time) was proposed by several studies [e.g., 8]. Paleomagnetism can significantly contribute to this debate, because paleomagnetic records have proven crucial to quantifying rotation and latitudinal displacement. Strik et al. [3] suggest that continents moved horizontally during the late Archean and that the drift rates were faster than in the Phanerozoic based on paleolatitudinal change from ca. 2.8 Ga flood basalts in the Pilbara craton.

The Archean Biosphere Drilling Program (ABDP) was designed for continental drilling at the northeastern Pilbara craton, Western Australia. The present study is focused on the ABDP#1 drilled core of the Marble Bar Chert Member, the Towers Formation, Pilbara craton, and our objectives are to reconstruct an early to late Archean APWP for the Pilbara craton and understand the geodynamic development during early Archean time.

2. Geological setting

The Pilbara craton comprises early to middle Archean granitoid rocks and greenstones, which unconformably underlie a late Archean to early Proterozoic succession of volcanic and sedimentary rocks [9,10]. The older granitoid rocks and greenstone unit are mainly distributed in the northern part of the Pilbara craton. Hickman [9] provided the first regional geological description of the older granitoid rocks and greenstones unit, and grouped the unit into a single stratigraphic sequence named the “Pilbara Supergroup.” In the eastern part of the Pilbara craton, the Pilbara Supergroup is subdivided into five volcanic and sedimentary groups and two additional formations: the Coonterunah Group, Warrawoona Group, Wyman Formation, Golden Cockatoo Formation, Sulphur Springs Group, Gorge Creek Group, and De Grey Group, in ascending order, ranging from 3.5 to 2.9 Ga in age [11].

In the Marble Bar Greenstone Belt, the survey area of the present study, the Warrawoona Group is mainly exposed (Fig. 1). Hickman [9] subdivided the Warrawoona Group into two major basaltic divisions, the lower Talga–Talga Subgroup, and the upper Salgash Subgroup (see Fig. 1 Legend). The Marble Bar Greenstone Belt of this region is overturned and steeply tilted to the east, and the strike is NNW–SSE. This geological feature is consistent with the ages of these groups and formations, being younger to the west. This deformation was mainly caused by an intrusion of the Mount Edgar Granitoid Complexes [e.g., 12,13] (Fig. 1). Van Kranendonk et al. [11] documented that the metamorphism of this region was also related to this granitoid intrusion. They reported that the grade of metamorphism changed from lower amphibolite facies to very low temperature metamorphism depending on the distance from the granitoid complexes. The age of deformation and metamorphism of the Marble Bar Greenstone Belt is constrained between 3320 and 3310 Ma, based on the age of pre-, syn-, and post-kinematic phases of the batholith [13,14]. The Towers Formation lies at the base of the Salgash Subgroup and is composed of volcanic rocks and intercalated chert layers. One of the major chert layers, the Marble Bar Chert Member, is of a low metamorphic grade. The thickness of the Marble Bar Chert Member is approximately 150 m in this study area. Other thin chert layers also occur in the Towers Formation with a total thickness of 26.4 m [7]. The volcanic rocks are named Apex Basalt. The age of the Marble Bar Chert Member within the Towers Formation is constrained by the underling Duffer Formation and the overlying Panorama Formation. The U–Pb zircon ages of the Duffer Formation are 3471±5 and 3465±3 Ma [15], and 3464±2 Ma [16]. The U–Pb zircon age of the Panorama Formation is 3458±1.9 Ma [15]. Thus, the depositional duration of the Marble Bar Chert Member is considered to be between 2.1 and 19.9 Myr. It is important to record that the ages used for this maximum and minimum depositional duration were obtained in same laboratory [15], and thus, the estimate of the depositional duration is appropriate.

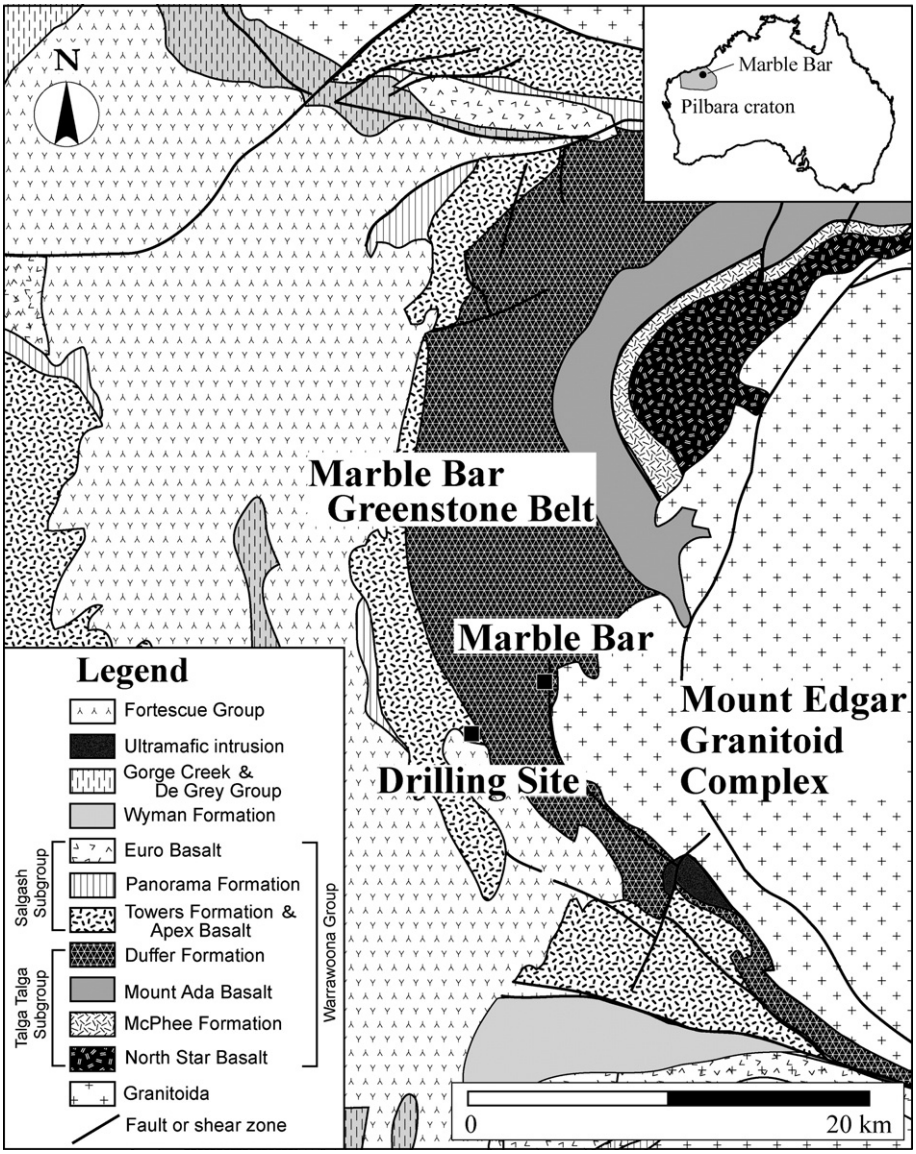


Fig. 1. Geological map of the eastern part of the Pilbara craton, after [11].

The lithofacies of the Marble Bar Chert Member on the outcrop are characterized by centimeter-scale beds of red, grey, white, and rare black layers. The Marble Bar Chert Member represents brecciation of the chert, cut by fault, with the breccia material showing disruption of the original structure. Oliver and Cawood [17] reported that this brecciation was related to the high pressure induced by the overturning of the strata. Kato and Nakamura [7] have studied the sedimentary processes of the Marble Bar Chert Member based on a geochemical signature. They argued that the positive correlation of P, V, Zn, and Y with Fe, and a positive Eu anomaly decrease (negative Ce decrease) as ∑REE and LREE/HREE increase, observed from the chert, is similar to the modern hydrothermal iron-rich sediment of mid-ocean ridges (MOR). Based on this result, they concluded that the Marble Bar Chert Member derived from a high temperature hydrothermal solution emanating from an MOR.

3. Sampling and material

In Australia, Tertiary lateritisation and intense weathering of surface rocks were reported in several studies [e.g., 18]. To avoid them, drilling of the strata is a powerful way to obtain fresh samples [e.g., 19]. In summer 2003, the ABDP drilled from 2.8 to 3.5 Ga Archean volcanic and

sedimentary units at five sites (#1–5) in the eastern Pilbara craton, Western Australia (summarized in [20]). The ABDP #1 core, which is analyzed in the present study, was drilled approximately 300 m southeast of the Marble Bar Pool, in the northeastern part of the Pilbara craton (21°12.57′ S, 119°43.92′ E). At the ABDP #1 drill hole, a continuous 270 m long core was obtained from the Towers Formation, including the Marble Bar Chert Member and the Apex Basalt. The orientation of the core was confirmed at 10 m intervals. The direction of the drill hole was a plunge angle of 40° toward 230°, oblique to the strike and dip of the strata, enabling the orientation of the cores. The strike and dip of the beds remained constant from the surface to the bottom of the drilled core from the Marble Bar Chert Member. The constant strike and dip were confirmed by a field survey of surface outcrop. The Marble Bar Chert Member is overturned, and strikes 323° and dips 83°.

The core is divided into volcanics and cherts corresponding to the Apex Basalt and the Marble Bar Chert Member (Fig. 2). The volcanic zone is subdivided into basalt and dolerite zones. The Marble Bar Chert Member is subdivided into three lithological zones: a white chert zone, an unstratified black chert zone, and a varicolored chert zone.

The varicolored chert zone represents intercalation structures of red, black, and white cherts, with a scale from a few to dozens of centimeters. Measurements of natural remanent magnetization (NRM) and magnetic susceptibility of all drilled cores indicate that only the red and black chert have a substantial magnetic intensity [21]. There are fragment quartz vein intrusions disturbing the upper part of the varicolored chert zone, while layered stratification is ambiguous from 140 to 158 m. Because of this, we analyzed the varicolored chert from 158.5 to 182.0 m of the ABDP#1 drilled core. Microscopic observations for this interval revealed that the red and black cherts represent clear lamination, indicating that primary structures of sedimentation have been preserved [22]. Thus, the varicolored chert was not affected by hydrothermal alteration, metamorphism, or modern weathering.

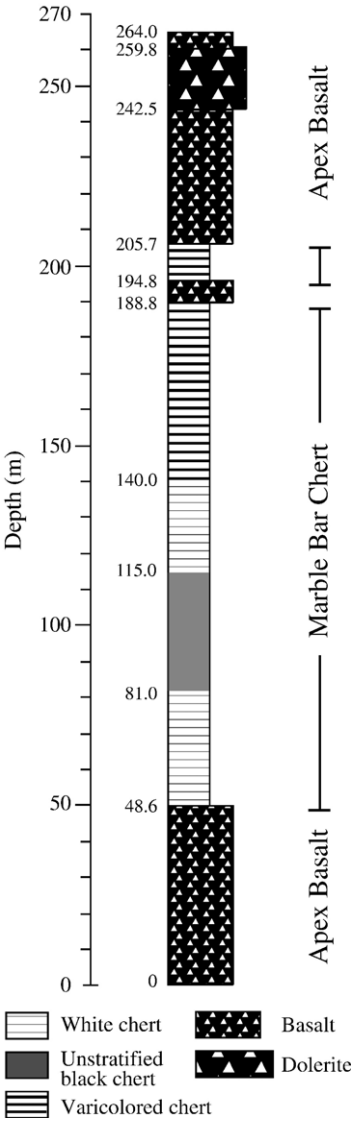


Fig. 2. Schematic stratigraphic profiles of drilled core of the Archean Biosphere Drilling Program (#1). Generalized lithology is shown in this figure. Chert and volcanics correspond to the Marble Bar Chert Member and the Apex Basalt, respectively.

4. Rock magnetism

4.1. Procedure

Progressive thermal demagnetization (ThD) of composite isothermal remanent magnetizations (IRMs) was conducted to identify the carrier of the remanent magnetization based on coercivity, intensity of the acquired IRM, and its blocking temperature. The three IRM components were acquired in direct magnetic fields of 50 (soft), 200 (medium), and 1000 mT (hard) and subsequent progressive ThD was conducted from100 °C to 500 °C with 50 °C intervals and from 500 °C to 690 °C with 30 °C–40 °C intervals [23]. Low-temperature phase transition of magnetic minerals, such as Verwey transition of magnetite at ~125 K was examined using a Quantum Design Magnetic Property Measurement System (MPMS).

4.2. Results

Results of the progressive ThD of composite IRMs are displayed in Fig. 3 for specimens taken from red and black

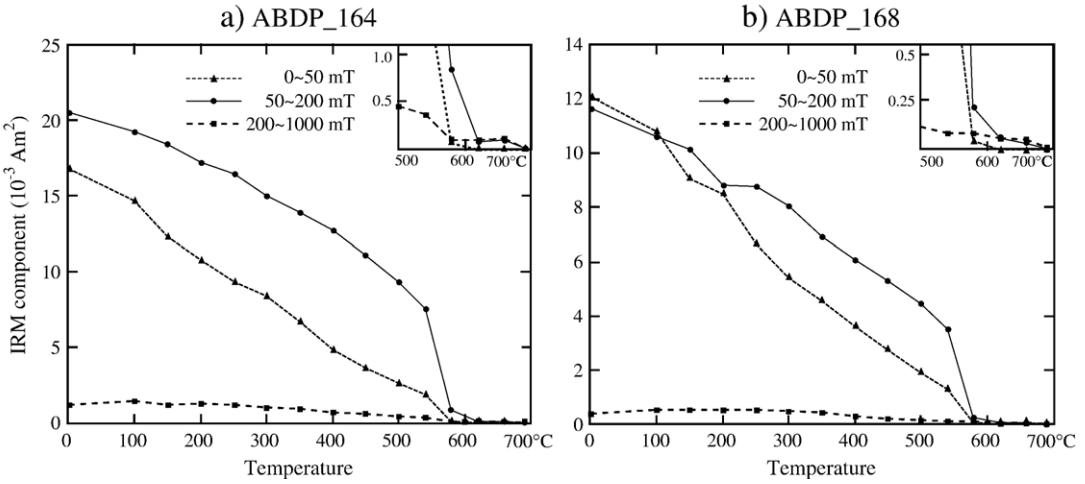


Fig. 3. Progressive thermal demagnetization of the three component IRMs (50, 200, 1000 mT). a) red chert and b) black chert.

chert, and they show two major drops of magnetization. The first drop appears on the 50 and the 200 mT components between 540 °C and 580 °C, indicating the existence of magnetic mineral with soft to medium coercivity, probably corresponding to magnetite or low-Ti titanomagnetite. The second drop is observed from 660 °C to 690 °C on the 1000 mT component, and slightly on the 200 mT component. This high coercivity and high blocking temperature mineral is probably hematite. In addition, the similarity of the results between Fig. 3a and b suggests that the red and black cherts have no difference in coercivity and blocking temperature.

Results of low temperature magnetometry for selected specimens are shown in Fig. 4. A large magnetization drop is observed at 125 K, which is considered to be the manifestation of the Verwey transition of magnetite. This result suggests that the specimens contain nearly pure magnetite, free from impurities or low temperature oxidation. This result is consistent with the result of the progressive ThD of composite IRMs.

5. Paleomagnetism

5.1. Procedure

Specimens for paleomagnetic analysis were cut into cubes of 1 cm^3, and were subjected to progressive alternative field demagnetization (AFD) and ThD. Alternating fields were from 10 mT to 60 mT with 5 mT intervals. Demagnetization temperatures were from 100 °C to

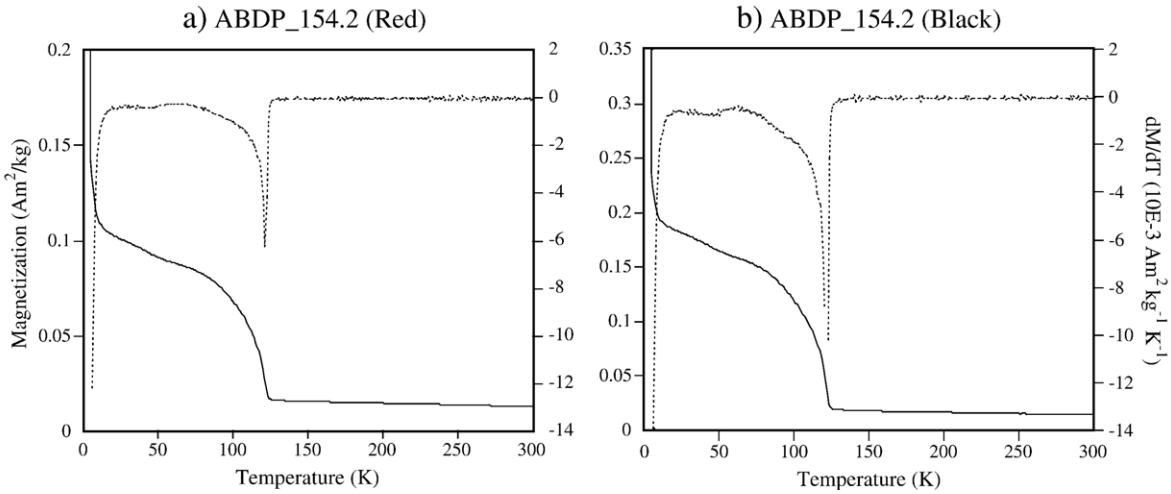


Fig. 4. Low temperature magnetometry for selected specimens. Solid lines (left axis) are the results of thermal demagnetization of field cooled SIRM. Dotted lines are the derivative of the curves (right axis).

500 °C with 50 °C intervals, and from 500 °C to 575 °C–600 °C with 25 °C intervals after 10 mT AFD treatment. Magnetic susceptibility was measured on selected specimens after each ThD step to detect the formation of secondary magnetic minerals or other thermal alterations. The results of progressive AFD and ThD are shown on the

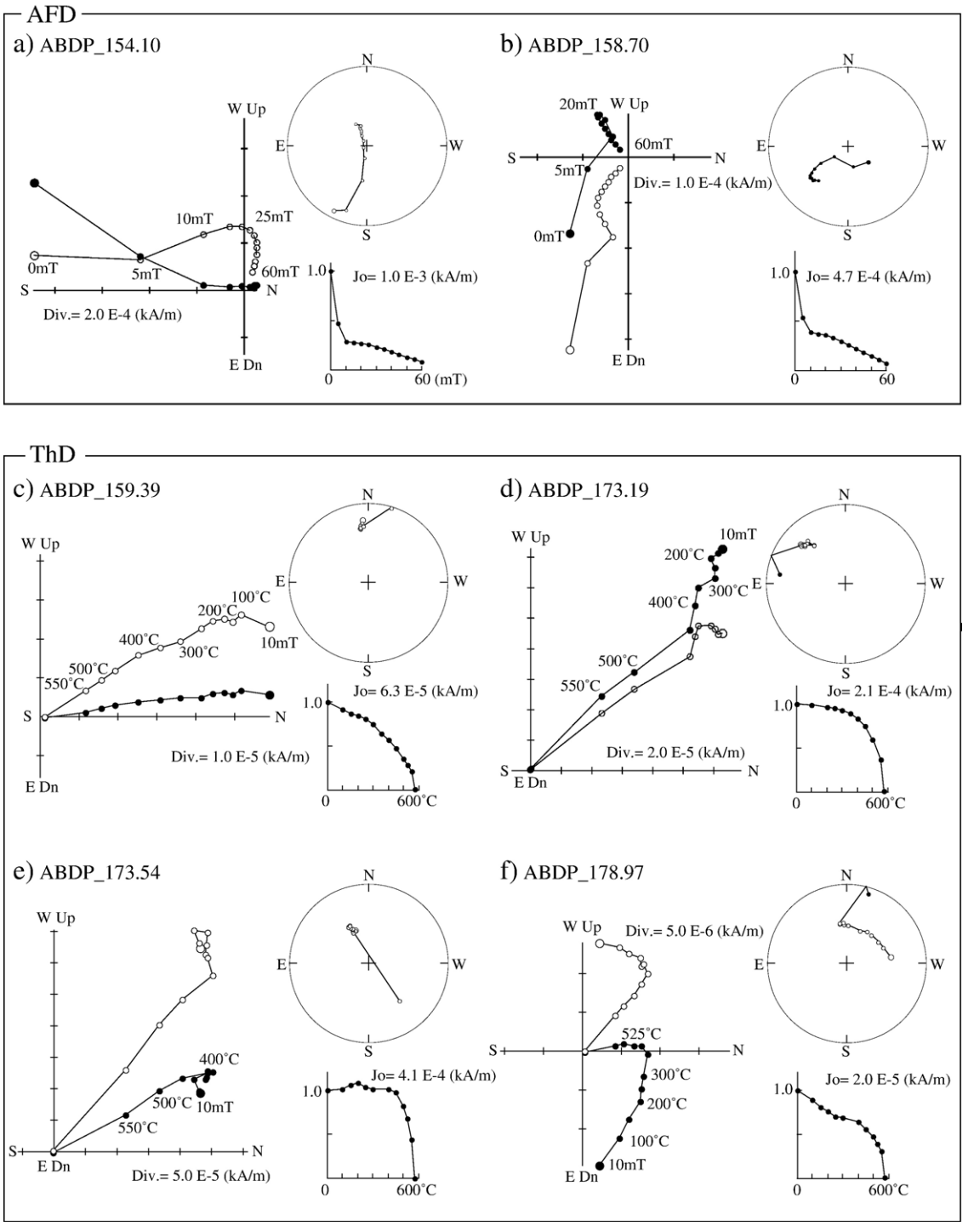


Fig. 5. Examples of progressive alternating field demagnetization (a, b) and thermal demagnetization (c–f) after tilt correction. For each sample we show the following. Left: an orthogonal vector plot with solid (open) symbols for data projected onto the horizontal (vertical) plane [24]. Upper right: equal area projection. Lower right: normalized intensity of magnetization versus demagnetization field or temperature.

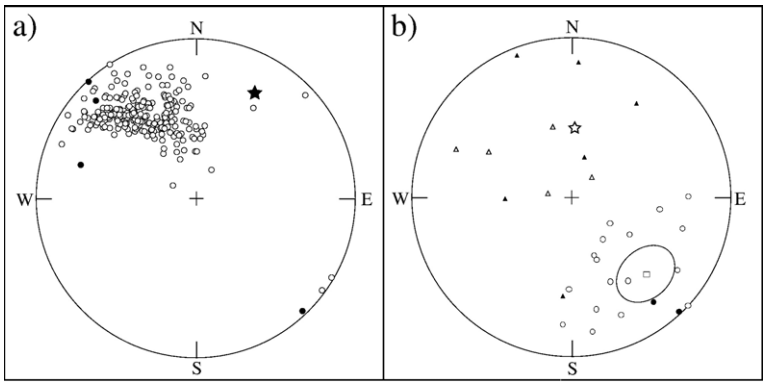


Fig. 6. Equal-area projection for paleomagnetic directions of the MT component after bedding correction (a) and in-situ LT components (b). Little circles are all accepted directions and star symbols represent the present Earth's magnetic field direction of this region. Open (solid) symbols indicate negative (positive) inclinations. In Fig. 6b, square symbols are site means with corresponding 95% confidence limit and small triangles in figure are rejected direction.

Zijderveld plot [24]. Characteristic Remanent Magnetization (ChRM) directions were calculated by principal component analysis [25]. Directions with a maximum angular deviation (MAD) less than 10° were employed for calculating site-mean directions using the Fisher statistics [26]. Remanent magnetizations were measured by a 2G 755R three-axis cryogenic magnetometer.

5.2. Results

A total of 431 discrete specimens were analyzed in the present study, and examples of AFD and ThD treatments are shown in Fig. 5. The AFD results show that more than 50% of NRM intensity is demagnetized by up to 10 mT, and the remaining magnetization is

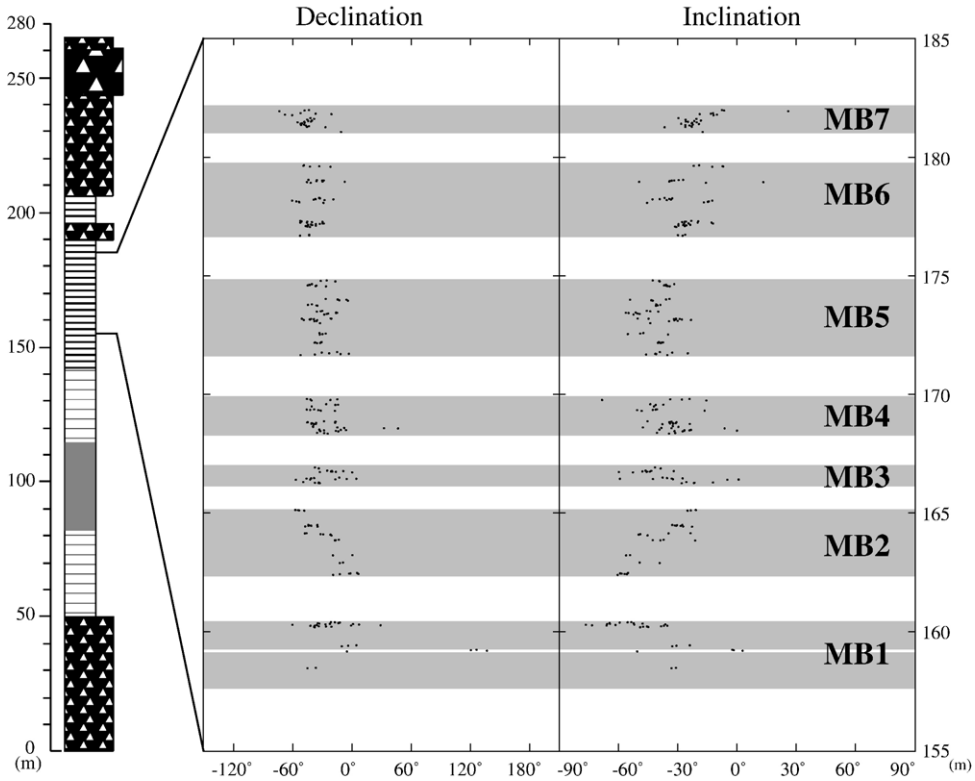


Fig. 7. Declination and inclination profile of the Marble Bar Chert Member of the drilled cores. MT component of each specimen is used for this column.

gradually demagnetized through the demagnetization steps that follow. This implies that there are two types of magnetite, each having different coercivity. An artificial overprint caused by drilling, which has been reported as a magnetic component in a direction vertical to a drill hole [e.g., 27], was adequately removed by up to 10 mT AFD.

Specimens generally hold either one or two magnetic components. Fig. 5a shows a large swing of magnetic direction through AFD, which indicates an overlap of magnetic coercivity spectra. In contrast, ThD is more effective in isolating the two magnetic components. A low temperature (LT) component was recorded in the temperature range of 100° to 250 °C–350 °C, and a moderate temperature (MT) component was recognized above 400 °C. (Fig. 5c–f). Fig. 5f shows two stage drops in magnetic intensity, with approximately 30% intensity loss between 100 °C and 300 °C indicating unblocking of the LT component and the MT component starting unblocking at temperatures higher than 400 °C.

The MT component was observed from 261 of the specimens (61%) by ThD treatment. The ChRM directions of the MT component after bedding correction are plotted in Fig. 6a. Most of them direct northwest upward, with a slightly elongated distribution from north to northwest in declination, and steeper to shallower in inclination. In order to assess the cause of this distribution, we plotted the declination and inclination of each ChRM direction with their stratigraphic sequence (Fig. 7). Because this record shows no stratigraphically uniform distribution, we divided them into seven stratigraphic windows as MB1 to MB7, and calculated mean paleomagnetic pole for each window (Table 1).

Stable ChRM directions of the LT component are observed from only 34 specimens. Their in-situ directions are plotted in Fig. 6, showing a loose clustering with southeast upward directions. We also calculated a mean paleomagnetic direction (MBL) for this grouping.

6. Discussion

6.1. Origin of the MT component and paleomagnetic direction

The mean paleomagnetic directions (MB1 to MB7) are significantly different from the present Earth's magnetic field (Fig. 6a). In order to evaluate the reliability of the MT component, virtual geomagnetic poles (VGPs) of MB1 to MB7 were compared with other published paleomagnetic poles in the Pilbara craton during early Archean time (Table 1). The pioneer study by McElhinny and Senanayake [2] analyzed samples from the Duffer Formation, which was dated as 3462–3476 Ma [15,16]. They collected paleomagnetic samples from columnar dacites and pillow basalts at two sites and compared their paleomagnetic directions. ThD treatment revealed that the remanent magnetization of the Duffer Formation consists of three components (LT, MT, and HT) differentiated by blocking temperatures. Based on the blocking temperature analysis and a positive fold test, they concluded that the MT component is primary magnetization carried by magnetite. Yoshihara [28] conducted a paleomagnetic study of the Apex Basalt, which underlies the Panorama Formation dated as 3458±1.9 Ma [15] and overlies the Marble Bar Chert Member. Paleomagnetic samples were collected from pillow basalts at five sites near the Marble Bar Pool. After ThD treatment, Yoshihara [28] found clustered paleomagnetic directions, which he regarded as a reliable primary magnetization of the Apex Basalt.

Table 1
Paleomagnetic poles from the Pilbara craton during early Archean

Rock unit	code	*n*	Dec	Inc	*k*	A95	P. lat	lat	long	Reference
Duffer Formation	DFM	11	335.5	31.8	41.0	7.2	17.2	43.9	86.3	McElhinny and Senanayake, 1980
Apex Basalt	AB	–	304.3	−36.8	–	5.0	−20.5	38.2	18.9	Yoshihara, 2001
Marble Bar Chert	MB1	27	337.4	−50.5	20.2	6.3	−31.3	67.4	−1.8	This present study
	MB2	28	333.9	−42.9	21.2	6.1	−25.0	65.7	15.4	This present study
	MB3	29	327.1	−34.6	16.3	6.8	−19.0	59.1	27.4	This present study
	MB4	44	333.1	−34.9	17.8	5.2	−19.2	64.7	28.9	This present study
	MB5	57	328.8	−40.4	48.4	2.7	−23.1	61.1	19.7	This present study
	MB6	48	319.9	−26.0	32.5	3.7	−13.7	51.1	33.7	This present study
	MB7	28	314.1	−19.6	25.9	5.5	−10.1	44.6	36.1	This present study

Paleomagnetic results from the core of the Marble Bar Chert Member with other published paleomagnetic poles from early Archean on the Pilbara craton. Aberrations; *n*, number of accepted specimens; Dec, declination; Inc, inclination; *k*, precision parameter; A95, angle of 95% confidence; P. lat, paleolatitude; lat, latitude of VGP; long, longitude of VGP.

The paleomagnetic poles of the Duffer Formation, the Apex Basalt, and MB1 to MB7 are presented in Fig. 8, and they show a whole continuous path of paleomagnetic poles from the Duffer Formation to the Apex Basalt.

6.2. Apparent polar wander path (APWP)

There are two Precambrian APWPs for the Pilbara craton [2,29]: however neither were well constrained during Archean time, because of scarcity of paleomagnetic poles. New paleomagnetic records obtained from the present study, along with recently published paleomagnetic records, provide a well constrained APWP during Archean time. McElhinny and Senanayake [2] concluded that the HT and the LT components corresponded to post-folding regional thermal events at 3.0 and 2.6 Ga, respectively. However these components lack precise age constraints of the remagnetization event. Hence, paleomagnetic poles of the MT component and Yoshihara's [28] paleomagnetic pole were used for constructing of the APWP during early Archean time. The Global Paleomagnetic Data Base (GPMDB) [30] contains no published paleomagnetic record from the Pilbara craton during middle Archean time. For the late Archean, several paleomagnetic poles have been reported [e.g., 3,31]. Strik et al. [3] and Strik [32] found a suite of paleomagnetic poles (P0–P10), which were confirmed by field tests as primary magnetization, using samples from the Fortescue Group in the Marble Bar region and the East Pilbara Basin, Pilbara craton.

Based on MB1 to MB7 and these published paleomagnetic poles, we constructed a synthetic APWP for the Pilbara craton from early to late Archean time (Fig. 8). The pole for the Duffer Formation [2] was taken as the starting point of the APWP.

6.3. Origin of the LT

The MBL, calculated from clustered directions of the LT component, provides a mean paleomagnetic direction of $D=136.1°$, $I=-33.0°$. Because the direction of the MBL is significantly away from the present Earth's magnetic field, the LT component can be interpreted as remagnetization acquired not recently but at a certain time after magnetization of the MT component. This remagnetization is likely to be related to a regional or local thermal event. To assess the origin of the remagnetization, both cases of normal polarity (MBL-n) and reverse polarity (MBL-r) of paleomagnetic poles are calculated from the

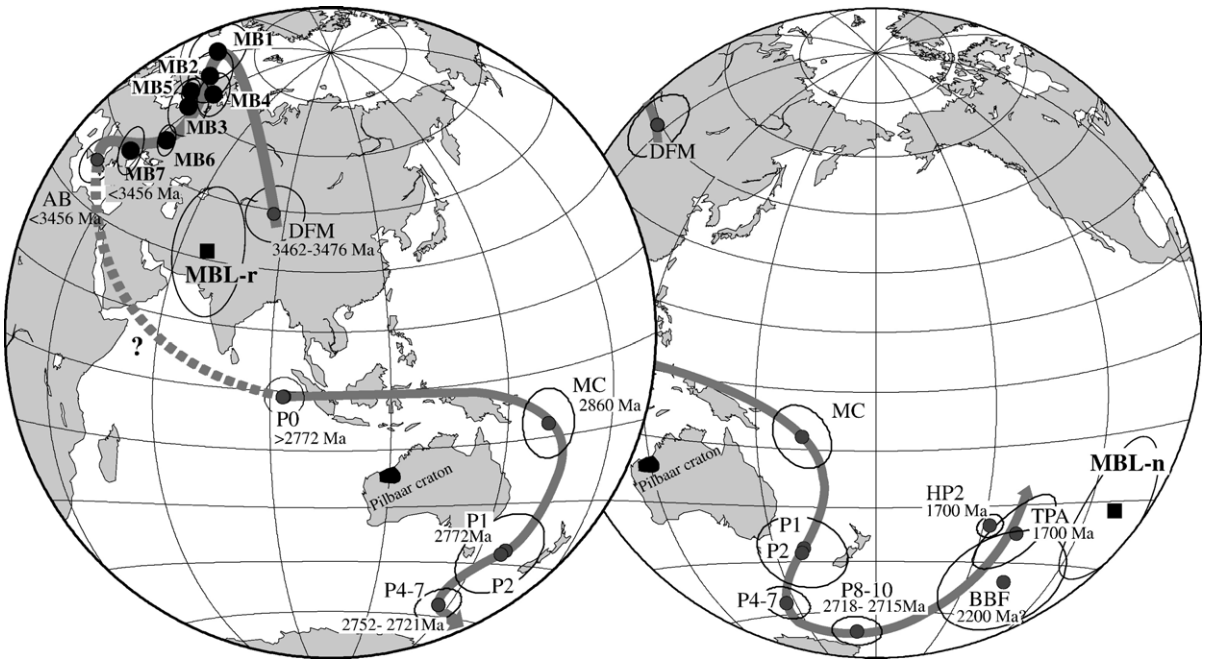


Fig. 8. Apparent polar wander path for the Pilbara craton during Archean time, shown in Lambert azimuthal equal area projections. Paleomagnetic poles of age-determined magnetization are displayed by circle symbols with corresponding 95% confidence limit. Square symbols represent paleomagnetic poles of age-undetermined magnetization. Abbreviations: MB1, MB2, MBL-n, and MBL-r (the present study); DFM [2]; AB [28]; MC [31]; P0–P10 [3,32]; BBF [34]; HP2 [35]; TPA [33]. This figure was drawn by the "PaleoMac" [42].

LT component and projected on the reconstructed APWP in Fig. 8.

The paleomagnetic pole of MBL-r appears to lie midway on the segment between AB to P0. Since the Marble Bar Greenstone belt experienced a metamorphism related to an intrusion of granitoid batholith at 3320–3310 Ma [13,14], this would indicate that the MBL-r component was acquired by this metamorphism between AB and P0. Another possibility is the remagnetization by a thermal event on the Pilbara craton at early Proterozoic. There are several reports which document burial metamorphism (TPA: [33]; BBF: [34]) or orogenic activity (HP2: [35]) at early Proterozoic. The paleomagnetic pole of the MBL-n is located in almost the same area as the paleomagnetic poles originating this thermal event (Fig. 8). Because the precision of paleomagnetic poles of the LT component is inadequate, we do not discuss further the origin of remagnetization in the present study.

6.4. Paleolatitude change

The newly reconstructed APWP reveals a significant change in the paleomagnetic pole position from MB1 to MB7. This corresponds to the change in paleolatitude being 21.2°, which means approximately 2350 km of lateral drift of the Pilbara craton. The depositional duration of the Marble Bar Chert Member (2.1–19.9 Myr) leads to an estimation of a lateral drift speed of 12–112 cm/yr (120–1120 km/Myr). Because only the latitudinal change is considered, this value is a minimum estimation. However, the change of paleomagnetic pole positions from MB1 to MB7 are observed only in a 29.6 m length of the core, whereas the total thickness of the Marble Bar Chert Member and other chert layers is approximately 176 m. Using linear interpolation, the depositional duration from MB1 to MB7 is estimated as 0.35–3.34 Myr, suggesting that a lateral drift speed may have attained 70.0–673 cm/yr (700–6730 km/Myr).

A possible explanation of the rapid lateral drift of the Pilbara craton is the plate tectonic process during early Archean time. The calculated lateral drift speed of 12–112 cm/yr (a much higher case is 70.0–673 cm/yr) between MB1 and MB7 suggests higher plate motion velocities compared with its current state [36]. Moreover, the maximum drift velocity is greater than the fastest velocity of plate motion ca. 30 cm/yr (300 km/Myr), during the last 200 Myr [37,38]. During late Neoproterozoic to Cambrian time, the final stage of supercontinental breakup and Gondwana assembly, the order of 20–40 cm/yr (200–400 km/Myr) of plate motion is reported by Meert [39]. He suggests that a reasonable explanation of the driving force for the anomalously high rates of plate motion is the lower mantle thermal anomalies. For late Archean time, Strik et al., [3] obtained reliable paleomagnetic records from ca. 2.8 Ga flood basalts in the Pilbara craton and reported 12–100 cm/yr (120–1000 km/Myr) of lateral drift. Their estimate of drift rates in late Archean time are almost equivalent to our estimation in early Archean time, suggesting that continents might have moved during the Archean faster than in the Phanerozoic.

On the other hand, a fundamentally different explanation is also possible to interpret the rapid lateral drift of the Pilbara craton, which is a true polar wander (TPW) hypothesis. TPW is a process by which the positive mass anomalies move toward the rotational equator. Evans [40] proposed that density anomalies in the context of whole mantle convection lead to TPW. However, the two-layer convection of the mantle during the Archean, proposed by Ida and Maruyama [41], seems to be inconsistent with large amounts of mass anomaly in the Earth's interior. The significant change of the paleomagnetic pole position may suggest, therefore, that the plate tectonics process has its roots in early Archean. To gain further insight, additional sampling of rocks with similar ages at different locations is required.

7. Conclusion

A paleomagnetic study, conducted on a continuous 270 m long drilled core from the Towers Formation, which includes the early Archean Marble Bar Chert Member in the Pilbara craton, revealed two distinct magnetic components (LT and MT). The MT component is divided into seven depth windows, which provide the mean paleomagnetic directions of MB1 to MB7. MB1 to MB7, coupled with previous paleomagnetic poles, display a continuous path of the paleomagnetic poles with age. This APWP implies a rapid shift of the paleomagnetic poles between MB1 and MB7, which corresponds to 21.2° in paleolatitude change, which is approximately 2350 km of latitudinal drift of the Pilbara craton. The depositional duration of the Marble Bar Chert Member yields an estimation of the lateral drift speed of 12–112 cm/yr (120–1120 km/Myr). The plate tectonic process would explain the fast lateral drift of the Pilbara craton. In this case, the maximum drift velocity is greater than the fastest velocity of plate motion ca. 30 cm/yr (300 km/Myr), during the last 200 Myr. This indicates that continents might have moved significantly faster during the Archean than in the Phanerozoic. Although, true polar wander (TPW) might instead explain the significant paleolatitude change, the two-layer

convection of the mantle during the Archean seems to be inconsistent with large amounts of mass anomaly in the Earth's interior to cause the large scale of TPW.

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